Asset page sections

OnCommand Insight

NetApp
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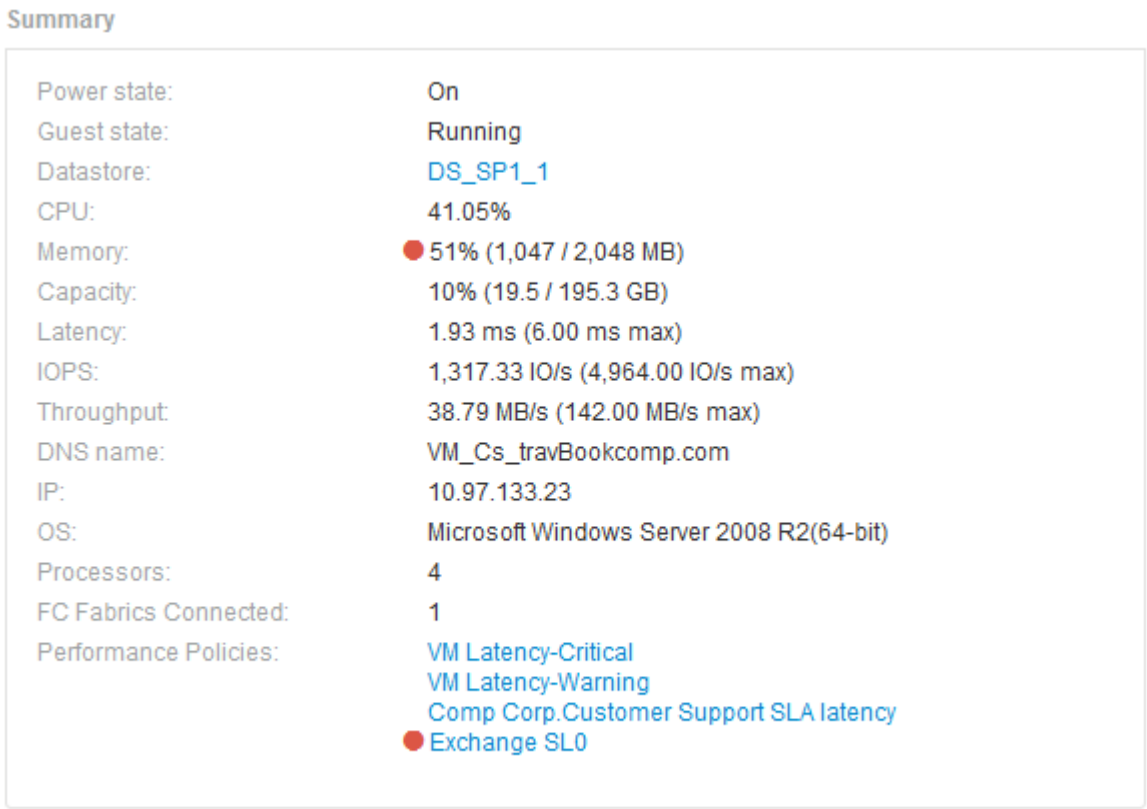
Asset page sections

An asset page displays several sections containing information relevant to the asset. The sections that you see depend on the type of asset.

Summary

The Summary section on an asset page displays a summary of information about the particular asset and shows issues related to the asset, indicated by a red circle, with hyperlinks to additional information about related assets and to any performance policies assigned to the asset.

The following example shows some of the types of information available in the Summary section of an asset page for a virtual machine. Any item with a solid red circle next to it indicates potential issues with the monitored environment.



Using the Summary section

You can view the Summary section to see general information about an asset. Specifically, it is helpful to see if any metrics (for example, memory, capacity, and latency) or any performance policies are cause for concern, which OnCommand Insight indicates by displaying a red circle next to the metric or performance policy.

Steps

1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays.

     The information that displays in the Summary section depends on the type of asset page you are viewing.
3. You can click any of the asset links to view their asset pages.

   For example, if you are viewing a storage node, you can click a link to view the asset page of the storage it is associated with or click to view the asset page of the HA partner.
4. You can view the metrics associated with the asset.

   A red circle next to a metric indicates that you might need to diagnose and resolve potential problems.

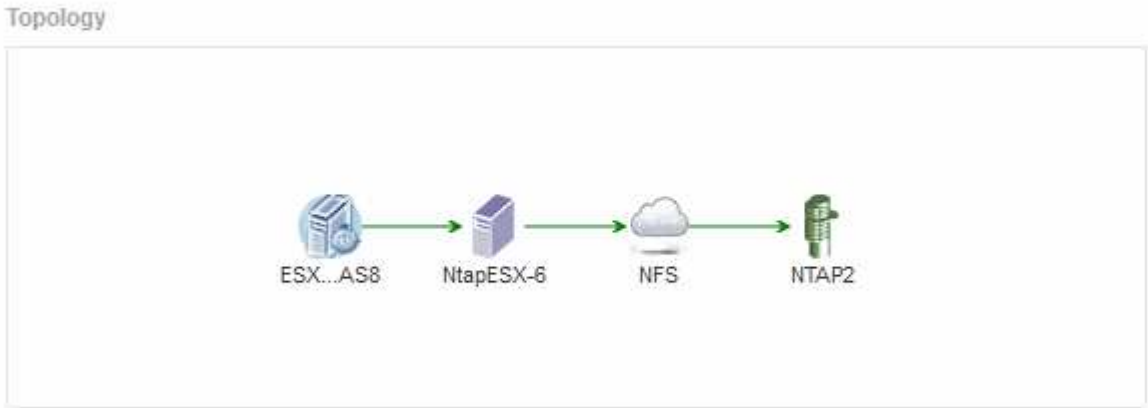
   You may notice that volume capacity might show greater than 100% on some storage assets. This is due to metadata related to the capacity of the volume being part of the consumed capacity data reported by the asset.
5. If applicable, you can click a performance policy link to view the performance policy or policies associated with the asset.

   If a red circle appears next to a performance policy, this indicates an asset has crossed the performance policy's defined threshold. You should examine the performance policy to further diagnose the issue.

Topology

The Topology section, if applicable to an asset, enables you to see how a base asset is connected to its related assets.

The following shows an example of what might display in the Topology section of a virtual machine asset page.



If the topology for the asset is larger than will fit in the section, the **Click link to see the topology** hyperlink is displayed instead.

Using the Topology section

The Topology section enables you to view how the assets in your network are connected to each other and display information about related assets.

Steps

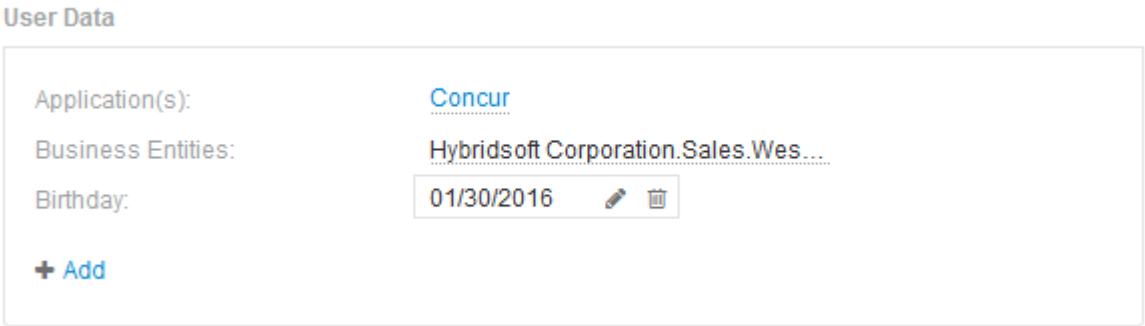
1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays. You can find the Topology section in the upper right-hand corner of the asset page.

   If the topology for the asset is larger than will fit in the section, click the **Click link to see the topology** hyperlink.
3. To view more information about the base asset's related assets, position your cursor over a related asset in the topology and click its name, which displays its asset page.

User Data

The User Data section of an asset page displays and enables you to change any user-defined data such as applications, business entities, and annotations.

The following shows an example of what might display in the User Data section of a virtual machine asset page when an application, business entity, and annotation are assigned to the asset:



Using the User Data section to assign or modify applications

You can assign applications running in your environment to certain assets (host, virtual machines, volumes, internal volumes, and hypervisors). The User Data section enables you to change the application assigned to an asset or assign an application or additional applications to an asset.

Steps

1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays.
3. You can do the following:
   - To view the asset page for the application, click the application's name.
   - To change the application assigned or to assign an application or additional applications, position your cursor over the application name, if an application is assigned, or over **None**, if no application is assigned, click , type to search for an application or select one from the list, and then click .

     If you choose an application that is associated with a business entity, the business entity is automatically assigned to the asset. In this case, when you place your cursor over the business entity name, the word *derived* displays. If you want to maintain the entity for only the asset and not the associated application, you can manually override the assignment of the application.
   - To remove an application, click .

Using the User Data section to assign or modify business entities

You can define business entities to track and report on your environment data at a more granular level. The User Data section in an asset page enables you to change the business entity assigned to an asset or remove a business entity from an asset.

Steps

1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays.
3. You can do the following:
   - To change the entity assigned or to assign an entity, click and select an entity from the list.
   - To remove a business entity, click .

> You cannot remove an entity that is derived from an application that is assigned to the asset.

Using the User Data section to assign or modify annotations

When customizing OnCommand Insight to track data for your corporate requirements, you can define specialized notes, called *annotations*, and assign them to your assets. The User Data section of an asset page displays annotations assigned to an asset and

also enables you to change the annotations assigned to that asset.

Steps

1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays.
3. In the **User Data** section of the asset page, click + Add.

   The Add Annotation dialog box displays.
4. Click **Annotation** and select an annotation from the list.
5. Click **Value** and do either of the following, depending on type of annotation you selected:
   - If the annotation type is list, date, or Boolean, select a value from the list.
   - If the annotation type is text, type a value.
6. Click **Save**.

   The annotation is assigned to the asset. You can later filter assets by annotation using a query.
7. If you want to change the value of the annotation after you assign it, click and select a different value.

   If the annotation is of list type for which the **Add values dynamically upon annotation assignment** option is selected, you can type to add a new value in addition to selecting an existing value.

Expert view

The Expert View section of an asset page enables you to view a performance sample for the base asset based on any number of applicable metrics in context with a chosen time period (3 hours, 24 hours, 3 days, 7 days, or a custom time period) in the performance chart and any assets related to it.

The following is an example of the Expert View section in a volume asset page:

You can select the metrics you want to view in the performance chart for the time period selected.

The Resources section shows the name of the base asset and the color representing the base asset in the performance chart. If the Top Correlated section does not contain an asset you want to view in the performance chart, you can use the **Search assets** box in the Additional resources section to locate the asset and add it to the performance chart. As you add resources, they appear in the Additional resources section.

Also shown in the Resources section, when applicable, are any assets related to the base asset in the following categories:

- Top correlated

  Shows the assets that have a high correlation (percentage) with one or more performance metrics to the base asset.

- Top contributors

  Shows the assets that contribute (percentage) to the base asset.

- Greedy

  Shows the assets that take away system resources from the asset through sharing the same resources, such as hosts, networks, and storage.

- Degraded

  Shows the assets that are depleted of system resources due to this asset.

Expert View metric definitions

The Expert View section of an asset page displays several metrics based on the time period selected for the asset. Each metric is displayed in its own performance chart. You can add or remove metrics and related assets from the charts depending on what data you want to see.

Metric	Description
BB credit zero Rx, Tx	Number of times the receive/transmit buffer-to-buffer credit count transitioned to zero during the sampling period. This metric represents the number of times the attached port had to stop transmitting because this port was out of credits to provide.
BB credit zero duration Tx	Time in milliseconds during which the transmit BB credit was zero during the sampling interval.
Cache hit ratio (Total, Read, Write) %	Percentage of requests that result in cache hits. The higher the number of hits versus accesses to the volume, the better is the performance. This column is empty for storage arrays that do not collect cache hit information.
Cache utilization (Total) %	Total percentage of cache requests that result in cache hits
Class 3 discards	Count of Fibre Channel Class 3 data transport discards.
CPU utilization (Total) %	Amount of actively used CPU resources, as a percentage of total available (over all virtual CPUs).
CRC error	Number of frames with invalid cyclic redundancy checks (CRCs) detected by the port during the sampling period
Frame rate	Transmit frame rate in frames per second (FPS)
Frame size average (Rx, Tx)	Ratio of traffic to frame size. This metric enables you to identify whether there are any overhead frames in the fabric.
Frame size too long	Count of Fibre Channel data transmission frames that are too long.
Frame size too short	Count of Fibre Channel data transmission frames that are too short.
I/O density (Total, Read, Write)	Number of IOPS divided by used capacity (as acquired from the most recent inventory poll of the data source) for the Volume, Internal Volume or Storage element. Measured in number of I/O operations per second per TB.

IOPS (Total, Read, Write)	Number of read/write I/O service requests passing through the I/O channel or a portion of that channel per unit of time (measured in I/O per sec)
IP throughput (Total, Read, Write)	Total: Aggregated rate at which IP data was transmitted and received in megabytes per second. Read: IP Throughput (Receive): Average rate at which IP data was received in megabytes per second. Write: IP Throughput (Transmit): Average rate at which IP data was transmitted in megabytes per second.
Latency (Total, Read, Write)	Latency (R&W): Rate at which data is read or written to the virtual machines in a fixed amount of time. The value is measured in megabytes per second. Latency: Average response time from the virtual machines in a data store. Top Latency: The highest response time from the virtual machines in a data store.
Link failure	Number of link failures detected by the port during the sampling period.
Link reset Rx, Tx	Number of receive or transmit link resets during the sampling period. This metric represents the number of link resets that were issued by the attached port to this port.
Memory utilization (Total) %	Threshold for the memory used by the host.
Partial R/W (Total) %	Total number of times that a read/write operation crosses a stripe boundary on any disk module in a RAID 5, RAID 1/0, or RAID 0 LUN Generally, stripe crossings are not beneficial, because each one requires an additional I/O. A low percentage indicates an efficient stripe element size and is an indication of improper alignment of a volume (or a NetApp LUN). For CLARiiON, this value is the number of stripe crossings divided by the total number of IOPS.
Port errors	Report of port errors over the sampling period/given time span.
Signal loss count	Number of signal loss errors. If a signal loss error occurs, there is no electrical connection, and a physical problem exists.

Swap rate (Total Rate, In rate, Out rate)	Rate at which memory is swapped in, out, or both from disk to active memory during the sampling period. This counter applies to virtual machines.
Sync loss count	Number of synchronization loss errors. If a synchronization loss error occurs, the hardware cannot make sense of the traffic or lock onto it. All the equipment might not be using the same data rate, or the optics or physical connections might be of poor quality. The port must resynchronize after each such error, which impacts system performance. Measured in KB/sec.
Throughput (Total, Read, Write)	Rate at which data is being transmitted, received, or both in a fixed amount of time in response to I/O service requests (measured in MB per sec).
Timeout discard frames - Tx	Count of discarded transmit frames caused by timeout.
Traffic rate (Total, Read, Write)	Traffic transmitted, received, or both received during the sampling period, in mebibytes per second.
Traffic utilization (Total, Read, Write)	Ratio of traffic received/transmitted/total to receive/transmit/total capacity, during the sampling period.
Utilization (Total, Read, Write) %	Percentage of available bandwidth used for transmission (Tx) and reception (Rx).
Write pending (Total)	Number of write I/O service requests that are pending.

Using the Expert View section

The Expert view section enables you to view performance charts for an asset based on any number of applicable metrics during a chosen time period, and to add related assets to compare and contrast asset and related asset performance over different time periods.

Steps

1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays. By default, the performance chart shows two metrics for time period selected for the asset page. For example, for a storage, the performance chart shows latency and total IOPS by default. The Resources section displays the resource name and an Additional resources section, which enables you

to search for assets. Depending on the asset, you might also see assets in the Top correlated, Top contributor, Greedy, and Degraded sections.

3. You can click **Select metrics to show**, and select a metric to add a performance chart for a metric.

   A performance chart is added for the selected metric. The chart displays the data for the selected time period. You can change the time period by clicking on another time period in the top left-hand corner of the asset page.

   You can perform the step again, and click to clear a metric. The performance chart for the metric is removed.

4. You can position your cursor over the chart and change the metric data that displays by clicking either of the following, depending on the asset:
   - **Read** or **Write**
   - **Tx**or **Rx Total** is the default.
5. You can drag your cursor over the data points in the chart to see how the value of the metric changes over the time period selected.
6. In the **Resources** section, you can do any of the following, if applicable, to add any related assets to the performance charts:
   - You can select a related asset in the Top correlated, Top contributors, Greedy, or Degraded sections to add data from that asset to the performance chart for each selected metric. Assets must have a minimum 15% correlation or contribution to be shown.

     After you select the asset, a color block appears next to the asset to denote the color of its data points in the chart.

   - For any asset shown, you can click the asset name to display its asset page, or you can click the percentage that the asset correlates or contributes to the base asset to view more information about the assets relation to the base asset.

     For example, clicking the linked percentage next to a top correlated asset displays an informational message comparing the type of correlation that asset has with the base asset.

   - If the Top correlated section does not contain an asset you want to display in a performance chart for comparison purposes, you can use the **Search assets** box in the Additional resources section to locate other assets. After you select an asset, it displays in the Additional resources section. When you no longer want to view information about the asset, click 🗑 .

Related Assets

If applicable, an asset page displays a Related Assets section. For example, a volume asset page might show information about assets like Storage Pools, Connected switch ports, and Compute Resources. Each section comprises a table that lists any of the related assets in that category, with links to their respective asset pages, and several performance statistics related to the asset.

Using the Related Assets section

The Related Assets section enables you to view any of the assets that are related to the base asset. Each related asset is displayed in a table along with pertinent statistics for

the asset. You can export the asset information, view the asset statistics in the Expert View performance charts, or show a chart that displays statistics for only related assets.

Steps

1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays.
3. To control how assets display in the table:
   - Click the name of any asset to display its asset page.
   - Use the **filter** box to show only specific assets.
   - Click a page number to browse through the assets by page if there are more than five assets in the table.
   - Change the sort order of the columns in a table to either ascending (up arrow) or descending (down arrow) by clicking the arrow in the column header.
   - Add a related asset to any performance chart in the Expert View section by placing your cursor over the related asset and clicking .
4. To export the information displayed in the table to a `.CSV` file:
   a. Click .
   b. Click **Open with** and then **OK** to open the file with Microsoft Excel and save the file to a specific location, or click **Save file** and then **OK** to save the file to your Downloads folder.

      All of the object attributes for the columns currently selected for display are exported to the file. Only the attributes for the displayed columns will be exported. Note that only the first 10,000 rows of the table are exported.
5. To display the related asset information in a chart below the table, click and do any of the following:
   - Click **Read**,**Write**, or **Total** to change the metric data that displays. **Total** is the default.
   - Click to select a different metric.
   - Click to change the chart type. **Line chart** is the default.
   - Move your cursor over the data points in the chart to see how the value of the metric changes over the time period selected for each related asset.
   - Click a related asset in the chart legend to add it to or remove it from the chart.
   - Click a page number in the related asset table to view other related assets in the chart.
   - Click to close the chart.

Violations

You can use the Violations section of an asset page to see the violations, if any, that occur in your environment as a result of a performance policy assigned to an asset. Performance policies monitor your network thresholds and enable you to detect a violation of a threshold immediately, identify the implication, and analyze the impact and

root cause of the problem in a manner that enables rapid and effective correction.

The following example shows aViolations section that displays on an asset page for a hypervisor:

Violations

filter...

Time	Description
06/05/2015 5:00:00 pm	Port balance index of 74 on esx1 exceeds the threshold of 50
06/12/2015 8:59:54 am	2 violations for esx2 with 'Swap out rate' > 3
06/12/2015 12:04:54 pm	esx1 violation with 'Swap out rate' > 3.00 KB/s (value of 86.85 KB/s)
06/12/2015 12:29:54 pm	esx1 violation with 'Swap in rate' > 3.00 KB/s (value of 59.90 KB/s)
06/12/2015 1:04:54 pm	7 violations for ds-30 with 'Latency - Total' > 50

Showing 1 to 5 of 32 entries

< 1 2 3 4 5 >

Using the Violations section

The Violations section enables you to view and manage any of the violations that occur in your network as the result of a performance policy assigned to an asset.

Steps

1. Log in to the OnCommand Insight web UI.
2. Locate an asset page by doing either of the following:
   - On the Insight toolbar, click , type in the name of the asset, and then select the asset from the list.
   - Click **Dashboards**, select **Assets Dashboard**, locate an asset name, and click it. The asset page displays. The Violations section displays the time the violation occurred and a description of the threshold that was crossed, along with a hyperlink to the asset on which the violation occurred (for example “2 violations fir ds-30 with Latency - Total > 50”).
3. You can perform any of the following optional tasks:
   - Use the **filter** box to show only specific violations.
   - Click a page number to browse through the violations by page if there are more than five violations in the table.
   - Change the sort order of the columns in a table to either ascending (up arrow) or descending (down arrow) by clicking the arrow in the column header.
   - Click the asset name in any description to display its asset page; a red circle indicates issues that need further investigation.

     You can click the performance policy, which displays the Edit Policy dialog box, to review the performance policy and make changes to the policy if necessary.
   - Click ✖ to remove a violation from the list if you determine the issue is no longer a cause for concern.

Customizable asset page

Additional data can be displayed in customizable widgets on each asset page. Customizing the page for an asset applies the customization to the pages for all assets of that type.

You customize asset page widgets by performing the following actions:

1. Add a widget to the page
2. Create a query or expression for the widget to showcase desired data
3. Choose a filter if desired
4. Choose a rollup or grouping method
5. Save the widget
6. Repeat for all desired widgets
7. Save the asset page

You can also add variables to the custom asset page that can be used to further refine your showcased data in widgets. In addition to regular variables, each asset type can use a set of "$this" variables to quickly identify resources directly related to the current asset, for example, all virtual machines hosted by the same hypervisor that hosts the current virtual machine.

This custom asset page is unique for each user as well as for each asset type. For example, if User A creates a custom asset page for a virtual machine, that custom page will display for any virtual machine asset page, for that user.

Users can only view, edit, or delete custom asset pages that they create.

Custom asset pages are not included in Insight's export/import functionality.

Understanding "$this" variables

Special variables on an asset's "Additional data" customizable page allow you to easily showcase additional information that is directly related to the current asset.

About this task

To use the "$this" variables in widgets on your asset's customizable landing page, follow the steps below. For this example, we will add a table widget.

> "$this" variables are only valid for an asset's customizable landing page. They are not available for other Insight dashboards. The available "$this" variables varies according to asset type.

Steps

1. Navigate to an asset page for an asset of your choosing. For this example, let's choose a Virtual Machine (VM) asset page. Query or search for a VM and click on the link to go to that VM's asset page.

   The asset page for the VM opens.

2. Click on the **Change view:** > **Additional Virtual Machine data** drop-down to go to that asset's customizable landing page.
3. Click on the **Widget** button and choose **Table widget**.

   The Table widget opens for editing. By default, all storages are shown in the table.

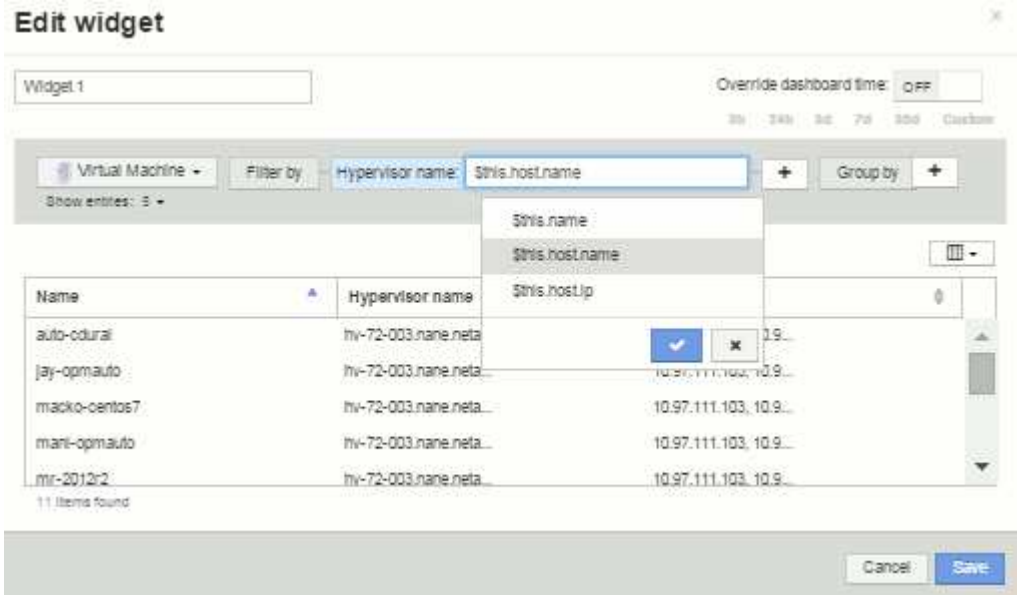
4. We want to show all virtual machines. Click on the asset selector and change **Storage** to **Virtual Machine**.

   All virtual machines are now shown in the table.

5. Click on the **Column selector** button and add the **hypervisor name** field to the table.

   The hypervisor name is shown for each VM in the table.

6. We only care about the hypervisor that hosts the current VM. Click on the **Filter by** field's**+**button and select **hypervisor name**.
7. Click on **Any** and select the **$this.host.name** variable. Click the check button to save the filter.



8. The table now shows all the VM's hosted by the current VM's hypervisor. Click **Save**.

Results

The table that you created for this virtual machine asset page will be displayed for any VM asset page you display. The use of the **$this.host.name** variable in the widget means that only the VM's owned by the current assets's hypervisor will be displayed in the table.

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